National University

of Computer & Emerging Sciences

Program: **BS (SE)**
Semester: **Fall 2026**
Target CLO: **CLO 1**
Course: **AI3002- Applied Machine Learning**

Assignment #**01**
Due Date: **10-10-2026**
Marks: **100**
Instructor: **Shoaib Khan**

Association Rule Mining–Based Recommendation System

1. Project Brief

In this group project, you will build a working recommendation system using Association Rule Mining / Market Basket Analysis. Your system must discover items that are frequently purchased together and use those discovered relationships to generate meaningful recommendations for a user.

The goal is not simply to run an Apriori library function. You must demonstrate the complete pipeline: **understand the transaction data, preprocess it, discover frequent itemsets, generate association rules, evaluate the rules, convert them into recommendations, and present the result through an interactive interface.**

2. Project Theme: Build Your Own Recommendation Experience

Each group must design and develop its own recommendation system application using Association Rule Mining. You are free to choose any suitable application domain that can naturally be represented using transaction or basket data.

Example Domains

- **Grocery / Supermarket:** Recommend products that are frequently purchased together (e.g., customers who buy bread and milk may also be recommended eggs).

- **Food Ordering:** Recommend complementary food items or add-ons (e.g., customers ordering pizza may also be recommended garlic bread or a soft drink).

Important: Unique Domain Allocation

- **First-Come, First-Served:** No two groups are allowed to work on the same application domain.
- **Confirmation Required:** Before beginning implementation, each group must select a domain and confirm that it is not already assigned to another group. Students must refer to **excel sheet provided in GCR** before finalizing their project domain.
- **Domain Conflicts:** If a domain has already been selected by another group, you must choose a different domain.

3. Mandatory Functional Requirements

A. Dataset

- Use a transaction/basket dataset containing multiple transactions and multiple items.
- Each transaction must have a transaction identifier and a set/list of purchased items.
- Document the dataset source, total number of transactions, number of unique items, and any preprocessing performed.

B. Data Preprocessing

- Handle missing, duplicate, inconsistent, or irrelevant records where applicable.
- Convert the raw data into a transaction representation suitable for association rule mining.

C. Frequent Itemset Mining

- Implement the Apriori workflow or use a standard Apriori implementation while demonstrating that the team understands its internal process.
- Generate frequent 1-itemsets, then proceed to larger candidate/frequent itemsets where possible.
- Apply minimum support and explain how changing this threshold affects the number of frequent itemsets.
- Demonstrate candidate generation and pruning conceptually within the report or interactive interface.

D. Association Rule Generation

- Generate rules from discovered frequent itemsets.
- Display antecedent (LHS), consequent (RHS), support, and confidence.
- Allow users to dynamically adjust minimum support and minimum confidence.
- Explain why A → B and B → A are distinct rules that can possess different confidence values.

E. Recommendation Engine

- Allow users to select or enter one or more items.
- Use discovered rules to generate recommendations for additional items.
- Prevent recommending items already selected by the user.
- Provide dynamic rationale for recommendations (e.g., 'Recommended because {bread, milk} → {eggs}').
- If multiple rules support a single recommendation, present the relevant evidence or combine/rank recommendations using a clearly documented method.

F. Evaluation

- Provide at least 5 meaningful recommendation examples.
- Manually inspect at least 10 discovered rules and classify them as useful, obvious/trivial, unexpected, or weak/uncertain.
- Discuss at least two limitations of association-rule-based recommendation engines.
- Include at least one experiment comparing system outcomes under different support and confidence thresholds.

4. Interactive Visual Interface (Required)

Build an interactive interface using Streamlit or a similar framework. Interface **SHOULD** include the following:

- **Dataset / Overview Page:** Displays total transaction count, unique-item count, most frequent items, and summary statistics.
- **Parameter Controls:** Interactive sliders/inputs for minimum support and minimum confidence.
- **Frequent Itemsets Page:** Displays generated frequent itemsets along with their corresponding support values.
- **Rules Explorer:** Displays antecedent, consequent, support, and confidence, with options for filtering and sorting.

- **Recommendation Page:** Interactive interface where users select items to receive real-time recommendations.
- **Explanation Panel:** Displays the specific rule(s) supporting each recommended item.
- **Visual Analytics:** Features at least two meaningful charts (e.g., item frequency distribution, support/confidence distributions, or network graph of association rules).

5. Must-Have Features

- **'Why This Recommendation?':** Explains every recommendation using the underlying rule and metrics.
- **Rule Search:** Allows users to search for rules containing specific items.
- **Threshold Playground:** Enables real-time observation of how rule sets expand or contract as support/confidence sliders adjust.
- **Smart Basket:** Simulates an interactive shopping basket where recommendations update automatically as items are added.
- **Recommendation Comparison:** Compares recommendations generated under two different threshold settings.
- **Association Graph:** Visualizes relationships with items as nodes and strong associations as connecting edges.
- **Natural-Language Explanation:** Converts rules into readable sentences (e.g., 'Customers who bought A and B also frequently bought C').
- **What-If Analysis:** Demonstrates how recommendations change dynamically when an item is removed from the shopping basket.

6. Required Mathematical Understanding

Metric / Term	Formula / Definition	Simple Interpretation
Support	support(A → B) = P(A ∩ B) = support({A, B})	How frequently A and B occur together across all transactions in the dataset.

Confidence	confidence(A → B) = P(B|A) = support(A ∩ B) / support(A)	The proportion of transactions containing A that also contain B.
Antecedent	Left-Hand Side (LHS)	The item(s) already selected or present in the user's basket.
Consequent	Right-Hand Side (RHS)	The item(s) recommended by the association rule.

7. Rule Analysis — Manual Inspection

After generating association rules from your project dataset, your group must manually inspect at least 10 discovered rules and classify each into one of the following categories (include this in your report):

Category	Definition
Useful	The rule provides actionable, meaningful, or commercially valuable insights.
Obvious / Trivial	The relationship is expected and provides little new operational value.
Unexpected	The relationship is surprising, non-obvious, or novel based on domain knowledge.
Weak / Uncertain	The rule has limited practical utility, low confidence, or insufficient evidence.

8. Recommended Technical Report Structure

- Introduction and Problem Statement
- Application Domain and Motivation

- Dataset Description
- Association Rule Mining Methodology
- Apriori Explanation and Hand-Worked Example
- Frequent Itemsets and Discovered Rules
- Recommendation Engine Logic
- Experiments with Support and Confidence Thresholds
- Results and Interpretation
- Manual Rule Evaluation (Useful, Trivial, Unexpected, Weak)
- Limitations

9. Suggested System Architecture

Your implementation should follow this end-to-end processing pipeline:

DATASET ➔ PREPROCESSING ➔ TRANSACTIONS ➔ APRIORI ENGINE ➔ FREQUENT ITEMSETS ➔ RULES ➔ RECOMMENDATION ENGINE ➔ STREAMLIT UI

10. Final Submission Checklist

Ensure all required components are completed and verified prior to submission:

- Source code repository / compressed package submitted with installation dependencies (requirements.txt).
- Runnable Streamlit application file (app.py or equivalent).
- Cleaned dataset file included or dataset download script provided.
- Technical Report (PDF/DOCX) structured per Section 8 guidelines.